Feature Engineering

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach a feature as an auditable transformation.

Supply all promised dimensions and comparison outputs,clarify reciprocal ratio names and preserve the association extension with its own evidence limits.

Keep one habit: Require formula,unit,time,state and same-case evaluation before claiming benefit.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Create features available at **prediction** time

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach two 45-minute sessions. This week focuses on **feature engineering**. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Evaluation values must not influence the learned **transformation**.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Saved classroom project records for critique; no personal student dataset required

Apply the connected idea: Relationships and confounding

Teach Week 21 as a two-session investigation using fictional data. In a fictional log, groups that practiced longer also scored higher: 1 hour with 4 points, 2 hours with 6 and 3 hours with 8. This pattern is an **association**.

Prior experience could affect both practice choices and scores. The table alone cannot establish the effect of adding practice time.

Assess the three stated outcomes through the independent task, not by copying the worked result.

Readiness answer

No; selection and coverage limit the claim.

Python 3 editor with a fresh local practice file

Paper trace table and the student workbook

Fictional **input** records from this lesson; no real student identifiers

Opening retrieval

Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

Look for: Read **training**-size learning curves

Reference example, not the only acceptable wording: Teacher explanation: Mean(5+7+6)/3=6; range 7-5=2.

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Record enough detail to reproduce a run

Reference example, not the only acceptable wording: Teacher explanation: Mean 70 percent; range 80-60=20 percentage points, across the three declared runs.

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

Look for: Freeze the full evaluation procedure before testing

Reference example, not the only acceptable wording: Teacher explanation: Accuracy 15/20, precision 3/5, recall 3/6. Report TN 12 FP 2 FN 3 TP 3 and blocked positive.

End goal

By the end, I can explain **feature engineering** using a traced example and a stated limitation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Learning Curves

Reproducible Experiments

Hold Out Test Data

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Length=12 cm, width=30 mm. Compute length/width using matching units. Explain handling for zero or missing width and where learned scaling is fitted.
</details>

Concept to explain

Feature engineering transforms raw **input**s into useful structure while preserving **prediction**-time **availability** and avoiding **leakage**.

Term	Meaning
transformation	rule converting raw values
derived feature	new input calculated from existing inputs
aspect ratio	width divided by length
feature pipeline	fixed ordered transformations

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach explicit feature formulas and training-only state

> Keep reciprocal ratios,main/pictured means and geometry/biology claims distinct.

Main availability and complete unit contract

A useful feature represents information available when the **prediction** is made. In a fictional parcel task, length and width are measured before packing, so area=length*width is available.

The final delivery delay is not available when predicting dispatch time. A strong correlation cannot make a future value legitimate.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **feature engineering**.

Feature step	Complete main dimensions / rule	Unit or boundary
Training rows	T1 length2, width 3; T2 length4, width 2; T3 length6, width 1, both dimensionscm	Areas6, 8, 6cm²
Fit centering state	Training mean area(6+8+6)/3=20/3	cm²; learned from training only
Validation V1	Length5 cm, width 2 cm→area 10cm²	Centered area 10-20/3=10/3cm²
Separate ratio example	Length12 cm, width 30 mm=3 cm→length_over_width4	Dimensionless ratio with declared order
Zero/missing width	Area formula permits measured 0 width→area 0; length/width cannot divide by0. Missing width is unobserved.	Return ratio unavailable for 0; request/reject missing dimensions under this contract.

Training rectangles have lengths [2, 4, 6] and widths [3, 2, 1], so areas are[6, 8, 6] square cm. **Training** mean area is 20/3. A validation rectangle length 5 width 2 has area 10 and centered area 10-20/3=10/3.

Freeze the **training** mean; do not recompute it with validation.

Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

Keep the distinction between an observed result and a proposed next step.

Term	Meaning
Feature engineering	Constructing model inputs from available measurements
Derived feature	An input computed from other inputs
Leakage	Information entering a model that would not be available legitimately
Availability	Whether a value exists when a prediction is needed

Feature engineering at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Raw measurements	Length and width	Available before prediction
Derived area	Multiply length by width	Units square cm
Centering	Subtract training mean	Reuse frozen value

Area is a deterministic transformation; it adds a representation, not a new measured fact. It may help a limited model express a relationship, or may do nothing useful. Compare original versus extended features under the same split, metric and development policy.

Center the validation area using the frozen training mean. Explain the steps or evidence behind your response.

Read separate pictured centering inputs

Engineer a feature with a **transformation** defined before evaluating a new leaf.

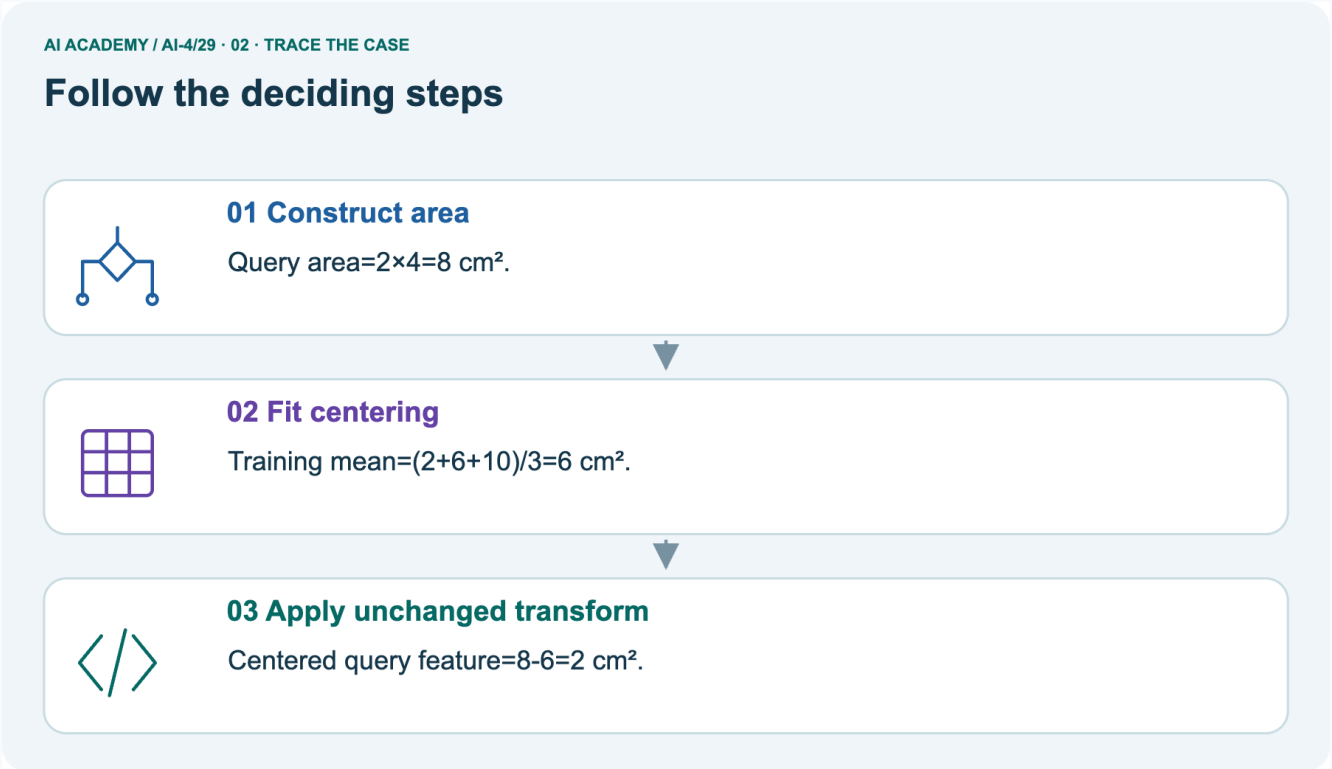
Training rectangular leaf cards have areas [2, 6, 10] cm². Area=length×width; center by **training** mean. Query length 2, width 4 cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Main training areas 6, 8, 6cm²give mean 20/3 and V1 area10 centered10/3. Picture is separate:training areas 2, 6, 10 give mean 6cm²; query 2×4=8cm²centers to2cm². Including that query in its mean gives(2+6+10+8)/4=6.5cm²and query 1.5cm², showing evaluation information changed fitted preprocessing. Centering preserves square-area units; it is not a unitless ratio.

These rectangle/bounding-card examples do not establish actual leaf area, disease diagnosis or feature benefit.

Trace unchanged mean and the query-in-mean defect



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Construct area	Query area=2×4=8 cm².
2. Fit centering	Training mean=(2+6+10)/3=6 cm².
3. Apply unchanged transform	Centered query feature=8-6=2 cm².

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. Centered query feature=8-6=2 cm². The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Area is a geometric toy feature. Useful predictive relationships need separate evaluation, and real leaves need a suitable measurement definition.

Change: Include the query in the centering mean.

Prediction: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

Reason: A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

▼ **Additional explicitly defined reciprocal ratio**

Earlier ratio example explicitly defines width_to_length=4/8=0.5 for length8cm,width4cm. The reciprocal length_over_width would be2. A ratio name alone is ambiguous;keep its numerator and denominator.

For this earlier width_to_length definition,4cm/8cm=0.5. Do not substitute the length_over_width formula used in the other exercise.

▼ Optional association/confounding explanation

In a fictional log, groups that practiced longer also scored higher: 1 hour with 4 points, 2 hours with 6 and 3 hours with 8. This pattern is an **association**. Prior experience could affect both practice choices and scores.

The table alone cannot establish the effect of adding practice time.

The numbers rise together in these three invented records. A description such as higher practice hours went with higher scores is supported. A claim that one extra hour guarantees two more points goes beyond the **observation**s.

No statistical calculation is required to see this limit.

Prior experience, motivation or available support could plausibly affect both. Explain the connection to each.

Exit: Small size, constructed data, conditions and measured outcome; do not generalize broadly or claim classroom effectiveness.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Create and justify two plant measurement features.

MODEL IDEA **Feature engineering** transforms raw **input**s into useful structure while preserving **prediction**-time **availability** and avoiding **leakage**.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 1 Raw leaf length=8 cm and width=4 cm.

STEP 3 Color histogram summarizes pixel ranges without using the target label.

STEP 4 Fit **transformation** settings on **training** data only; document unit, purpose, **availability**, and **leakage** check.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Features are leaf length and width in cm; add length/width.

Compute for8 and2; state boundary checks.

Exact explanation to model

Ratio=4, unitless with matching units. Guard zero/missing width; fit learned transforms on **training** only.

Explain the added worked example

Key terms

Apply the connected idea: Relationships and confounding

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

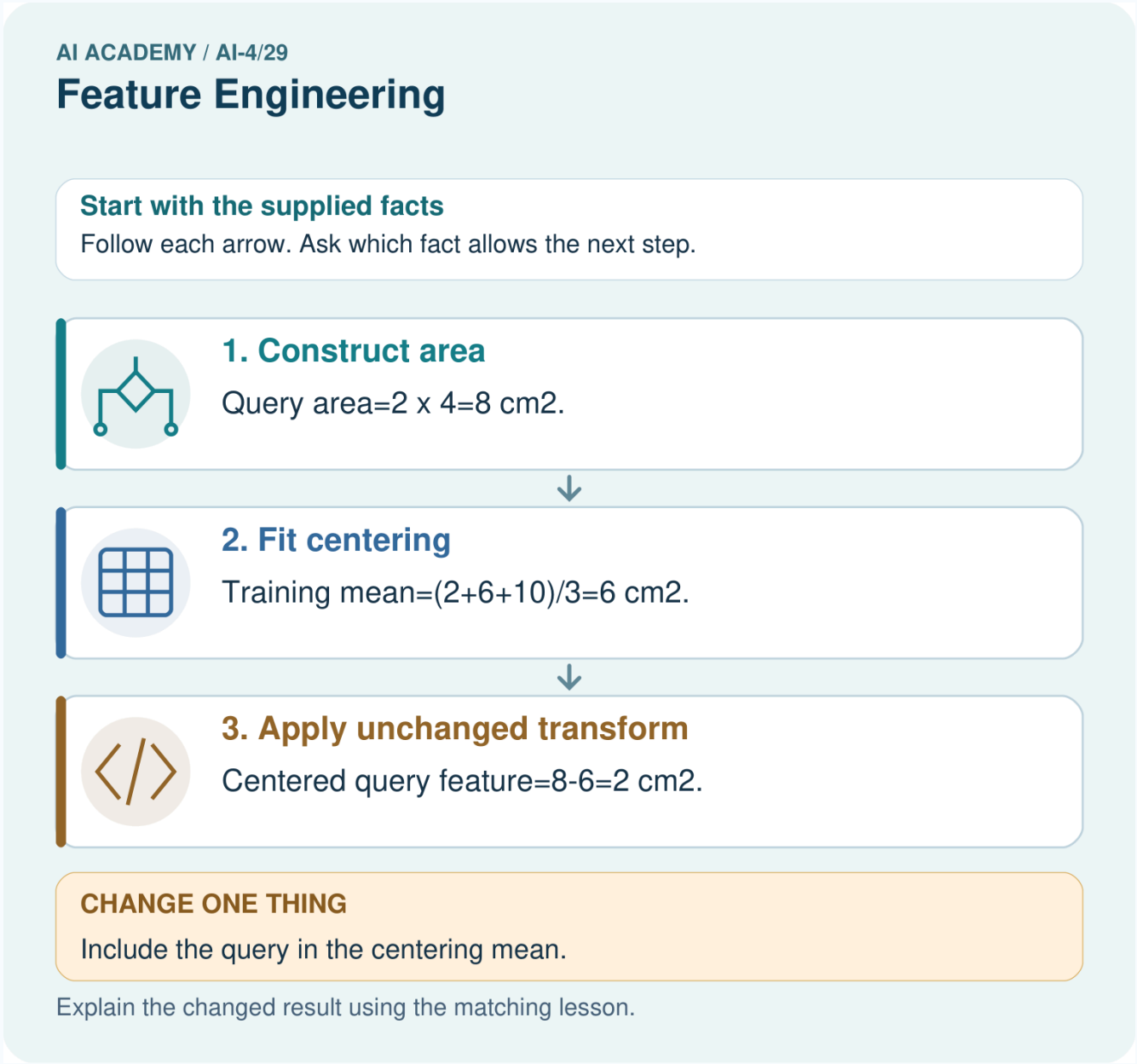
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Feature Engineering

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

Current completion task and checked key

Completion task: Center the validation area using the frozen **training** mean.

Teacher check: Teacher explanation: Area 10; centered 10-20/3=10/3 square cm. Validation does not change the fitted mean.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Feature engineering**. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: Areas 6, 8, 6 square cm; **training** mean 20/3 square cm.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

The area definition, cm² units and **training** mean 6; the correct declared centered feature is 2 cm². Changed-case check: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

Changed-case reasoning: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**. A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: Features are leaf length and width in cm; add length/width.

Write the question to explain: Compute for8 and2; state boundary checks.

Reveal after **prediction**s: Ratio=4, unitless with matching units. Guard zero/missing width; fit learned transforms on **training** only.

Use the earlier transfer question as additional practice. Reserve the selected fresh training-area4/8/12and3-by-4query task for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided response: Area 10; centered 10-20/3=10/3 square cm. Validation does not change the fitted mean.

Exit response: Compare with and without area on the same validation records, **fitting** all **transformation**s on **training** and reporting the declared metric and feature **availability**.

Case-specific teaching sequence

Stage · Teacher action

> **Stage:** Session 1 start to 7 minutes
>
> **Teacher action:** Use the prerequisite question: Why must preprocessing be fitted on training only? Check the explanation before proceeding.

Stage: 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify feature engineering in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: compute features; transform validation. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Connect each conclusion to the stated evidence.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between feature engineering and derived feature. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Why is delivery delay invalid for a prediction made before dispatch?

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Create features available at prediction time; Fit feature transformations using training evidence only. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Teach the connected idea

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve the prerequisite and explain the purpose in the opening section.
7 to 20 minutes	Read the worked example and diagram; ask learners to trace each step aloud.
20 to 33 minutes	Complete W1 and W2, then compare predictions with observed results.
33 to 45 minutes	Collect the guided prediction before running; discuss a mismatch and its cause.
Session 2 start to 8 minutes	Retrieve the main rule without showing the previous answer.
8 to 20 minutes	Use W3 to diagnose and repair the stated failure; rerun an earlier case.
20 to 35 minutes	Collect W4 independently, including code or plan, predictions and explanations, before hints.
35 to 45 minutes	Discuss W5 and the exit; use the fresh retry when support was needed.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

▼ Extension teaching cycles

Connected teaching cycle: Relationships and confounding

Target: Describe **association**s without claiming **cause**

Reserve further sessions for this topic; the 135-minute primary plan does not absorb all added content.

Completion practice: Explain how prior experience could relate to both practice time and score.

Teacher check: Experienced learners may choose different practice amounts and also score higher even without additional practice.

Individual check: For Relationships and confounding, use its worked case to evaluate this proposed explanation: “A successful run proves the conclusion”. Explain the deciding step.

Expected correction: A fairer **comparison** could offer the same task, instructions and assessment and randomly assign fictional groups to practice conditions.

In real classrooms, a teacher must approve any activity and protect equal learning opportunities; this lesson uses a paper plan.

Apply the same support, independent transfer and delayed-review decisions to the named connected-topic workbook tasks.

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Main area/centering, unit and missing policies as supplied; ratio numerator/denominator must be named explicitly. Source feature utility fixtureboth 3/4 differenterrors, not an executed training pipeline. Optional association hours 1/2/3, scores 4/6/8 doesnotprovepracticecausedgain; prior experience may confound. Independent trainingmean 8cm², queryarea 12cm², centered 4cm²; train-only fit andsquareunits required.

No model outcome supplied for that arithmetic.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.

2. 2. Feature Workshop: design two **transformation**s, compute examples, audit **leakage**, and compare one controlled validation result.

3. 3. Run one normal, one boundary, one missing, and one shifted case.

4. 4. Calculate the metric from raw counts or row-level errors.

5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: Areas 6, 8, 6 square cm; **training** mean 20/3 square cm.

A stronger claim makes a stronger project: Compare with and without area on the same validation records, **fitting** all **transformation**s on **training** and reporting the declared metric and feature **availability**.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: **Training** mean 8 square cm; new area 12 square cm; centered feature 4 square cm.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Extension misconceptions

A successful run proves the conclusion: A fairer **comparison** could offer the same task, instructions and assessment and randomly assign fictional groups to practice conditions.

In real classrooms, a teacher must approve any activity and protect equal learning opportunities; this lesson uses a paper plan.

Matching the example is enough: Require the new W4 case, explicit **prediction**s and an explanation. Fresh retry: a log links more library visits with higher reading scores. Reading interest may influence both.

Require a supported **association** statement and a fairer **comparison** plan.

A convenient shortcut preserves meaning: A chart title Practice **cause**s success overstates the **evidence**. Replace it with Practice time and scores in three fictional records, and describe the possible **confounder**.

Better wording does not repair the study, but it makes the report honest.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the selected fresh training-area4/8/12and3-by-4query task for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR Creating many features automatically gives the model more useful information.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Check these misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: **Training** areas [4, 8, 12]. A new rectangle has length 3 width 4. Compute the **training** mean and centered new feature; state units.

Teacher explanation: **Training** mean 8 square cm; new area 12 square cm; centered feature 4 square cm.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Compute features

Compute the worked areas and their **training** mean.

Teacher explanation: Areas 6, 8, 6 square cm; **training** mean 20/3 square cm.

W2 Transform validation

Center the validation area using the frozen **training** mean.

Teacher explanation: Area 10; centered 10-20/3=10/3 square cm. Validation does not change the fitted mean.

W3 Reject leakage

Why is delivery delay invalid for a **prediction** made before dispatch?

Teacher explanation: The value is only known later, so using it would not match the information available at **prediction** time.

W5 Plan a fair comparison

How would you test adding area without claiming it must help?

Teacher explanation: Compare with and without area on the same validation records, **fitting** all **transformation**s on **training** and reporting the declared metric and feature **availability**.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Ratio=4, unitless with matching units. Guard zero/missing width; fit learned transforms on **training** only.

Additional transfer practice question

A feature is leaf_length divided by leaf_width. A row has width zero. What should the pre**process**ing rule do before division?

Reasoned practice answer

Detect zero or invalid width, investigate or handle it using a documented rule, and avoid division by zero. Apply pre**process**ing consistently without learning from test data.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A feature formula is safe for every row just be**cause** it works on typical examples.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Width=3 cm, ratio=4 and unitless. Zero width makes division undefined; missing width is not zero. Use a documented validation/fallback policy.

Fit learned scaling only on **training** data and apply it unchanged to other partitions.

Explanations for the additional workbook examples

W1 Compute features

W2 Transform validation

W3 Reject leakage

W4 Apply independently

Training areas [4, 8, 12]. A new rectangle has length 3 width 4. Compute the **training** mean and centered new feature; state units.

Teacher explanation: **Training** mean 8 square cm; new area 12 square cm; centered feature 4 square cm.

W5 Plan a fair comparison

Assess independent transfer

Selected case answer

Feedback that the learner uses

▼ Extension questions and checked keys

Relationships and confounding — W1 Describe the evidence

State the pattern in the three worked records without using **cause**d or guaranteed.

Higher recorded practice hours are associated with higher scores in these three records.

Relationships and confounding — W2 Identify a confounder

Explain how prior experience could relate to both practice time and score.

Experienced learners may choose different practice amounts and also score higher even without additional practice.

Relationships and confounding — W3 Repair an overclaim

A learner predicts every student gains 2 points per extra hour. Explain why that is unsupported.

The three **observation**s do not establish a universal causal rule or predict every individual; other factors and variation are unmeasured.

Relationships and confounding — W4 Plan independently

A fictional dataset links more hints used with lower scores. Give a noncausal explanation and plan a fairer **comparison** of whether hints help.

Learners finding tasks harder may use more hints and score lower.

Use comparable tasks and starting skill, a teacher-approved or paper random-assignment plan, same outcome measure and report uncertainty; do not deny needed support.

Relationships and confounding — W5 State a remaining limit

Even a better **comparison** has only four fictional groups. What should be reported?

Small size, constructed data, conditions and measured outcome; do not generalize broadly or claim classroom effectiveness.

MAIN LESSON · SECTION 6

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Add only justified Plant Classifier features, with formula, source fields, expected benefit, and **leakage** test.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **feature engineering** and **derived feature** in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: **Training** mean 8 square cm; new area 12 square cm; centered feature 4 square cm. Fresh retry **training** areas [3, 6, 9], new rectangle 2 by 5: **training** mean 6, new area 10, centered 4 square units.

Require **training**-only **fitting**. Mark each criterion as independent, supported or not yet demonstrated. Reassess the reasoning after feedback; do not replace the first attempt silently.

Delayed return

Suggested return points in this level: ai-4/30, ai-4/32, ai-4/35. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 30: Improve Generalization. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 30: Improve Generalization. Carry forward feature formulas,units,prediction-time availability,training-fitted state and controlled same-case comparisons. Experiment records and seeds were introduced in Week 26.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

▼ Extension support and reassessment

Readiness and reassessment for Relationships and confounding

Draw arrows from the proposed **confounder** to both observed variables. Extension: explain why reversing **cause** and effect can also fit some **observation**al patterns.

Independent W4 **evidence**: Learners finding tasks harder may use more hints and score lower.

Use comparable tasks and starting skill, a teacher-approved or paper random-assignment plan, same outcome measure and report uncertainty; do not deny needed support.

Record whether the learner needed no help, a hint or explanation. Fresh retry: a log links more library visits with higher reading scores. Reading interest may influence both.

Require a supported **association** statement and a fairer **comparison** plan.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Create features available at **prediction** time

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Center the validation area using the frozen **training** mean.

Reasoned practice answer: Teacher explanation: Area 10; centered 10-20/3=10/3 square cm. Validation does not change the fitted mean.

Check every learner before discussion: Use the worked case for **Feature engineering**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: Areas 6, 8, 6 square cm; **training** mean 20/3 square cm.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Include the query in the centering mean. Require a **prediction** and a reason before comparing. Expected result: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

Reason: A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

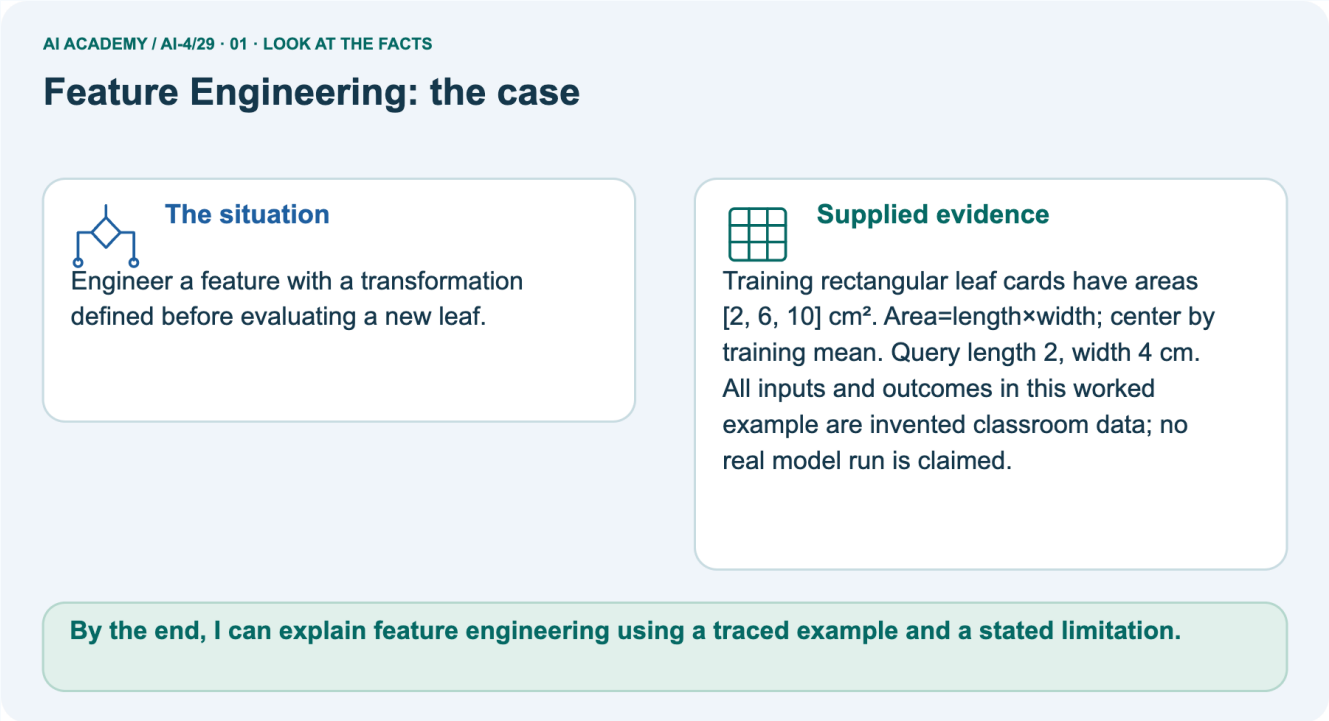
<details>
<summary>Reference · Teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Feature Engineering: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

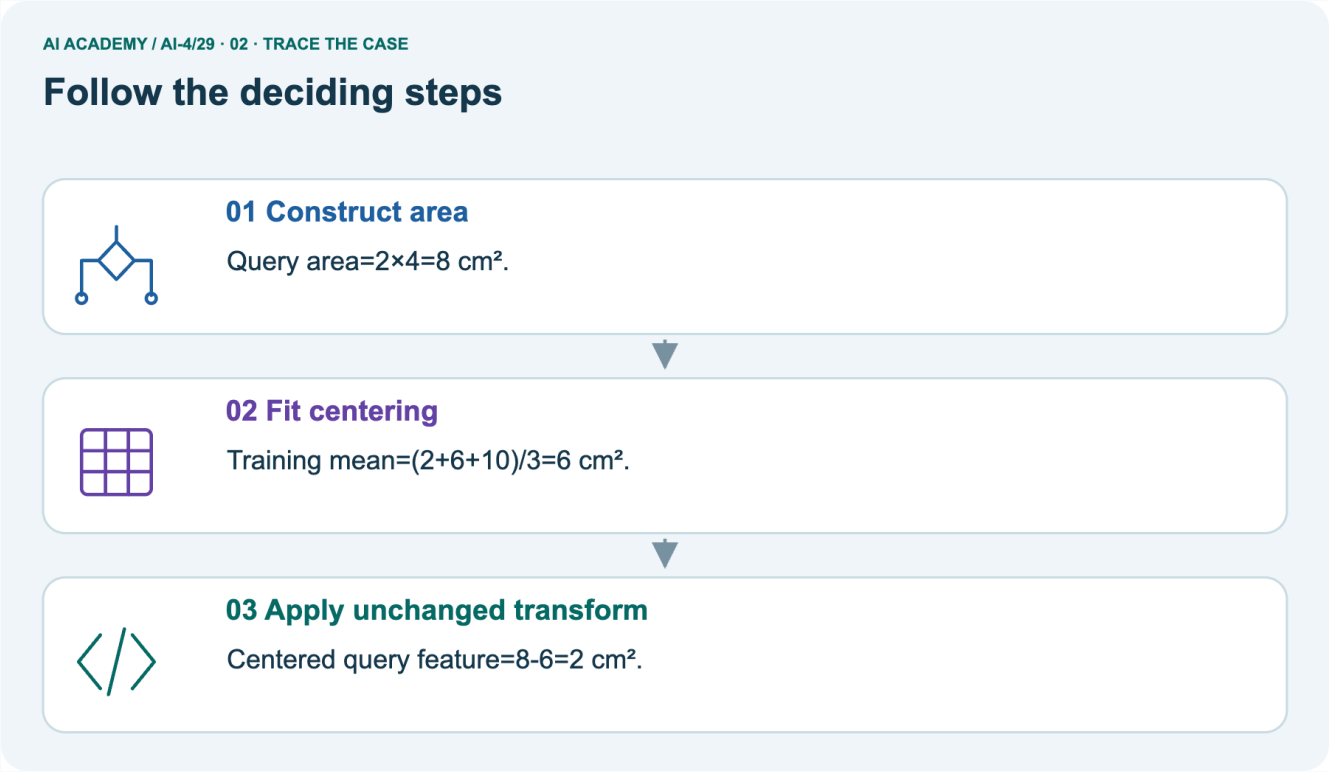
Engineer a feature with a **transformation** defined before evaluating a new leaf. Supplied facts: **Training** rectangular leaf cards have areas [2, 6, 10] cm². Area=length×width; center by **training** mean. Query length 2, width 4 cm.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



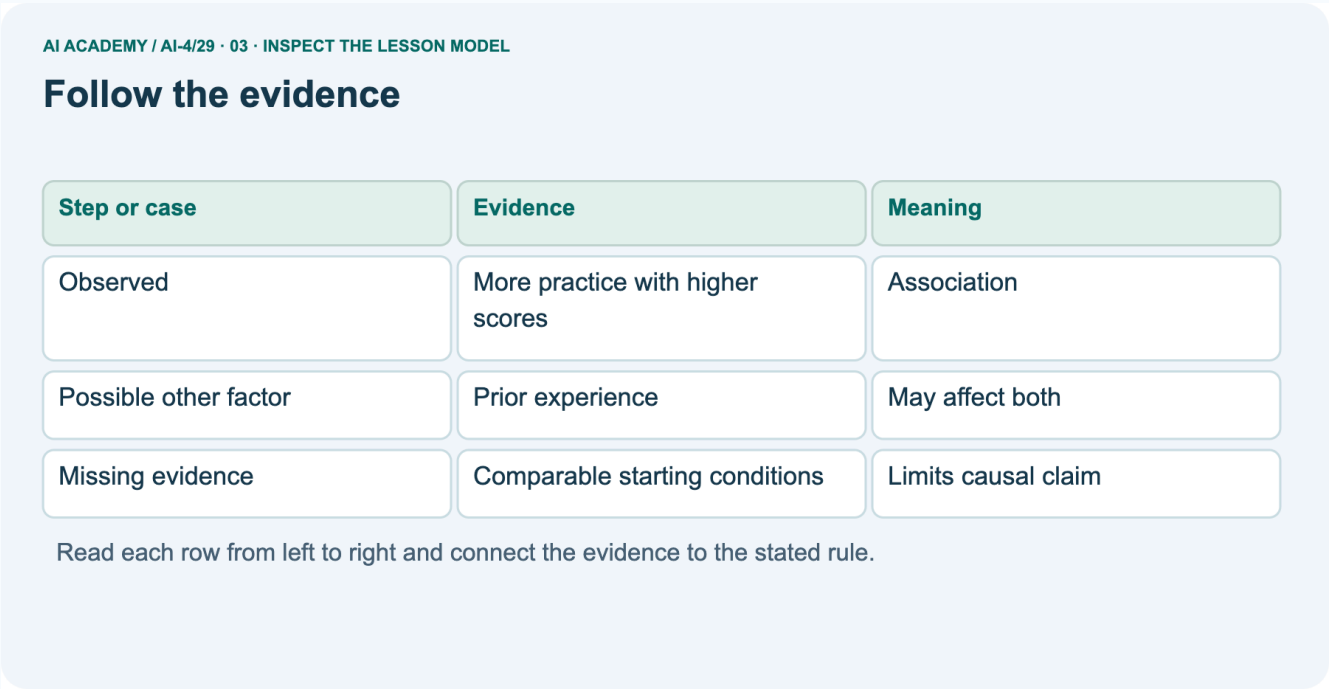
Follow the labelled picture with the matching explanation. Open the vector diagram

Construct area: Query area=2×4=8 cm². → Fit centering: **Training** mean=(2+6+10)/3=6 cm². → Apply unchanged transform: Centered query feature=8-6=2 cm².

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Follow the evidence



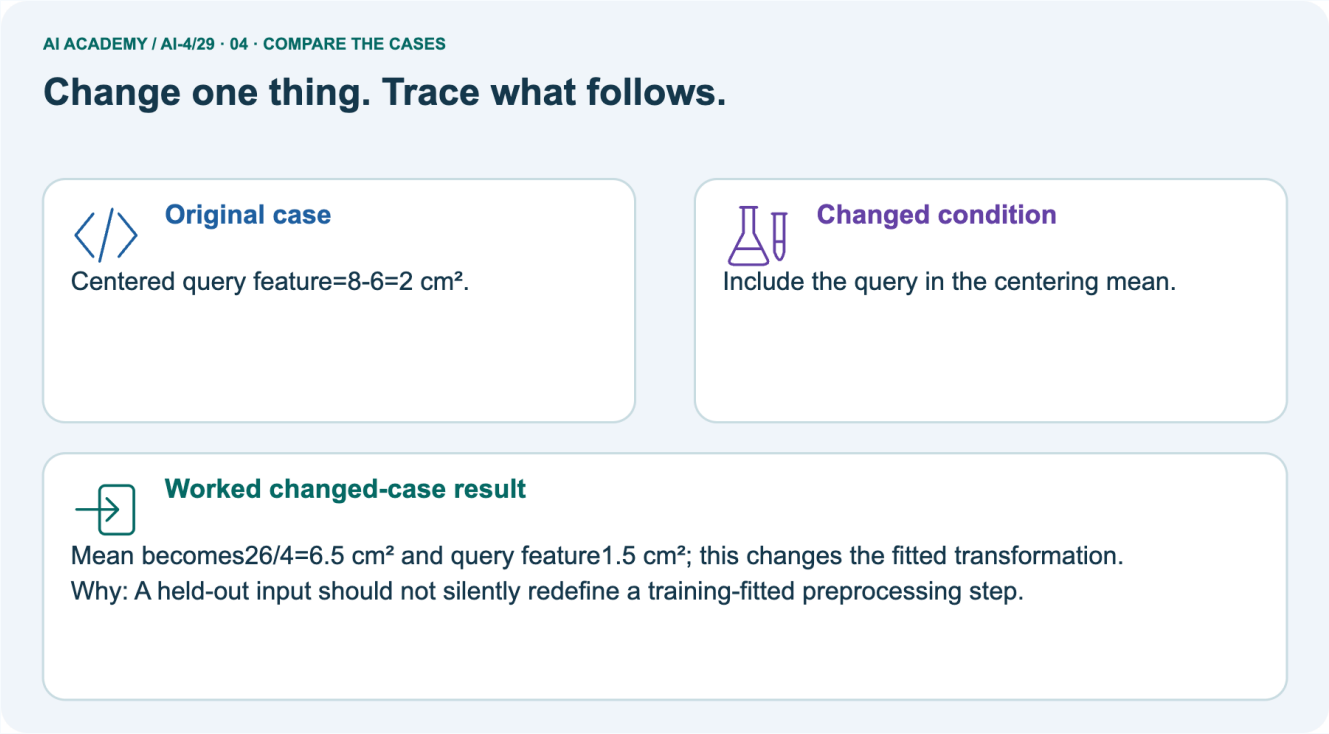
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Observed | More practice with higher scores | **Association**. Possible other factor | Prior experience | May affect both. Missing **evidence** | Comparable starting conditions | Limits causal claim

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Centered query feature=8-6=2 cm². Changed condition: Include the query in the centering mean. Expected result: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**.

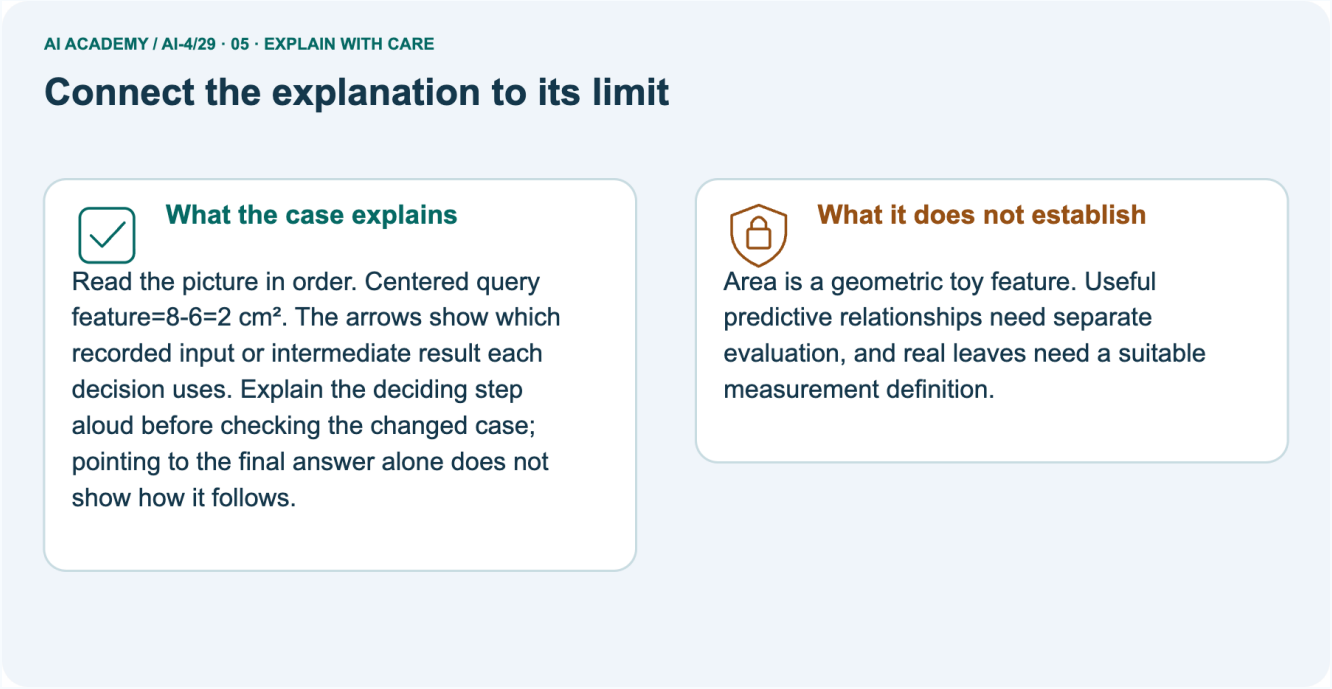
Reason: A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Centered query feature=8-6=2 cm². The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Area is a geometric toy feature.

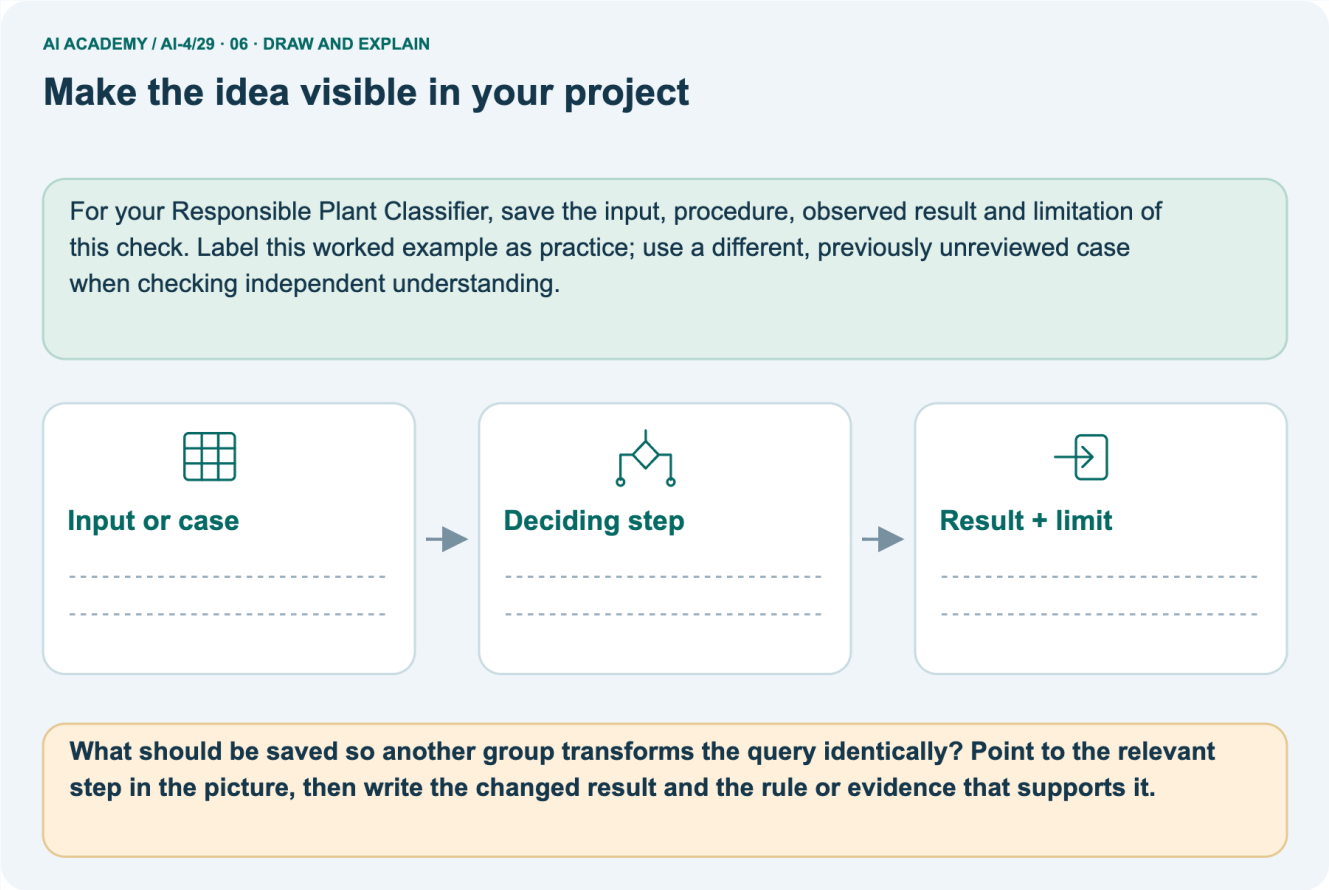
Useful predictive relationships need separate evaluation, and real leaves need a suitable measurement definition.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Area is a geometric toy feature. Useful predictive relationships need separate evaluation, and real leaves need a suitable measurement definition.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should be saved so another group transforms the query identically? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The area definition, cm² units and **training** mean 6; the correct declared centered feature is 2 cm².

Changed-case check: Mean becomes26/4=6.5 cm² and query feature1.5 cm²; this changes the fitted **transformation**. A held-out **input** should not silently redefine a **training**-fitted pre**process**ing step.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Feature Engineering: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.